

AMBUD LAHIRI

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EDUCATION

Bachelor of Technology in Computer Science (Artificial Intelligence)

Bennett University

August 2022 – Present

CGPA : 9.45

EXPERIENCE

Undergraduate Research Intern

Bennett University

August 2024 – May 2025

- Engineered a hypervisor-level cache side-channel TLB attack detection system using Intel's PCM, achieving 85% accuracy in identifying Flush+Reload, Flush+Flush, and Prime+Probe attacks through real-time monitoring of IPC.
- Added Explainable AI using SHAP to analyze feature importance by leveraging GradientExplainer.

PROJECTS

Kine-sense | Streamlit, Django, Python, SQLite, Daphne, Youtube API

April 2025 – May 2025

- Implemented a Python-centric video analytics platform with Django and Streamlit, reducing video loading times by 60 ms and improving the user experience for video engagement data.
- Deployed a real-time predictive analytics pipeline using Django Channels, enabling live inference on user engagement data, and subsequently improving potential revenue prediction accuracy by 15%.
- Leveraged FFmpeg and the YouTube Data API to build a data ingestion pipeline that automatically categorized 95% of incoming video data by content type, improving data organization.

Bail Reckoner | Next.js, Django, Python, Llama 3.1, Fast API

November 2024 – December 2024

- Developed and integrated 90% of core UI components, encompassing forms with immediate feedback, real-time data visualizations, and live updates, leading to smoother team workflows and higher user satisfaction.
- Facilitated seamless data flow by developing 30% of FastAPI endpoints, integrating the frontend with an AI-driven legal database; this reduced manual data entry errors by approximately 10% for legal professionals.
- Enhanced the integration of 55% of real-time legal database access and AI-driven precedent matching, improving automated case law retrieval and strengthening judicial decision-making accuracy.

BU-GPT | Flutter, Dart, OpenAI API, Firebase

August 2024 – September 2024

- Developed a Flutter-based mobile application integrated with the OpenAI API, providing interactive conversational AI and real-time document query functionalities, resulting in a 90% reduction in time spent on manual document analysis.
- Fortified mobile app security via Firebase Authentication, constructing an intuitive tab view navigation that streamlined user onboarding, decreasing the time spent by new users by 70%.
- Leveraged RESTful APIs and optimized asynchronous request handling using Dart's async/await, delivering responsive AI-generated insights, thereby improving user productivity by up to 95%.
- Revolutionized user experience through intuitive UI/UX design, leading to a 60% increase in daily active users and a 90% reduction in user-reported accessibility issues.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, NoSQL(MongoDB), Kotlin, Dart, JavaScript, Typescript, HTML, CSS.

Frameworks: Flutter, Streamlit, React.js, Next.js, Node.js, Express.js, Tailwind CSS.

Developer Tools: Git, GitBash, VS Code, Visual Studio, PyCharm, IntelliJ, Android Studio, Firebase, Tkinter.

CERTIFICATIONS

- Algorithmic Toolbox
- Supervised Machine Learning
- Software Development Processes and Methodologies